# End to End Supply Chain Analysis

A comprehensive diagnostic review of 172,765 orders revealing systemic shipping failures and actionable solutions to protect $2.1M in profit

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# Executive Summary

This analysis examined 172,765 orders processed between 2015 and 2018, uncovering a critical operational failure: **54.71% of all orders are delivered late**. The investigation identifies a systemic misalignment in shipping logic, particularly affecting premium shipping modes. Whilst the business generated $7.5M in profit during this period, the persistent delay rate currently puts $2.1M at risk of erosion.

Our analysis deployed a Random Forest predictive model that can now identify high-risk orders with **74% accuracy before they leave the warehouse**, providing an immediate tool for operational intervention. The findings reveal that the problem is not isolated to specific warehouses or customer segments, but represents a company-wide systemic flaw requiring immediate strategic action.

## Key Finding

First Class shipping has a 100% delay rate due to impossible promised windows, not poor performance

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# Business Context & Objectives

The company operates a global e-commerce platform specialising in sporting goods, apparel, and electronics. The analysis was commissioned to address a critical operational failure eroding customer trust and profitability.

### Core Problem

Actual shipping times consistently exceed promised delivery windows, leading to customer dissatisfaction and profit erosion

### Financial Impact

Quantify the monetary value at risk from delayed deliveries across all order categories

### Bottleneck Detection

Identify high-level and granular operational bottlenecks across six dimensions of the supply chain

### Predictive Capability

Develop a machine learning model to flag potential delays at the point of order placement

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# Critical Performance Metrics

**172,765**

**Orders Analyzed**

Total order volume across 4 years

**54.71%**

**Late Delivery Rate**

Over half of all orders delayed

**$2.1M**

**Profit at Risk**

Revenue threatened by delays

**3.0**

**90th Percentile Delay**

Days beyond promised window

**79%**

**Model Precision**

Accuracy identifying late risk

The data reveals that whilst 80.7% of orders remain profitable, the 18.7% of loss-making orders are disproportionately concentrated amongst delayed shipments. This indicates that operational friction is directly tied to margin erosion.

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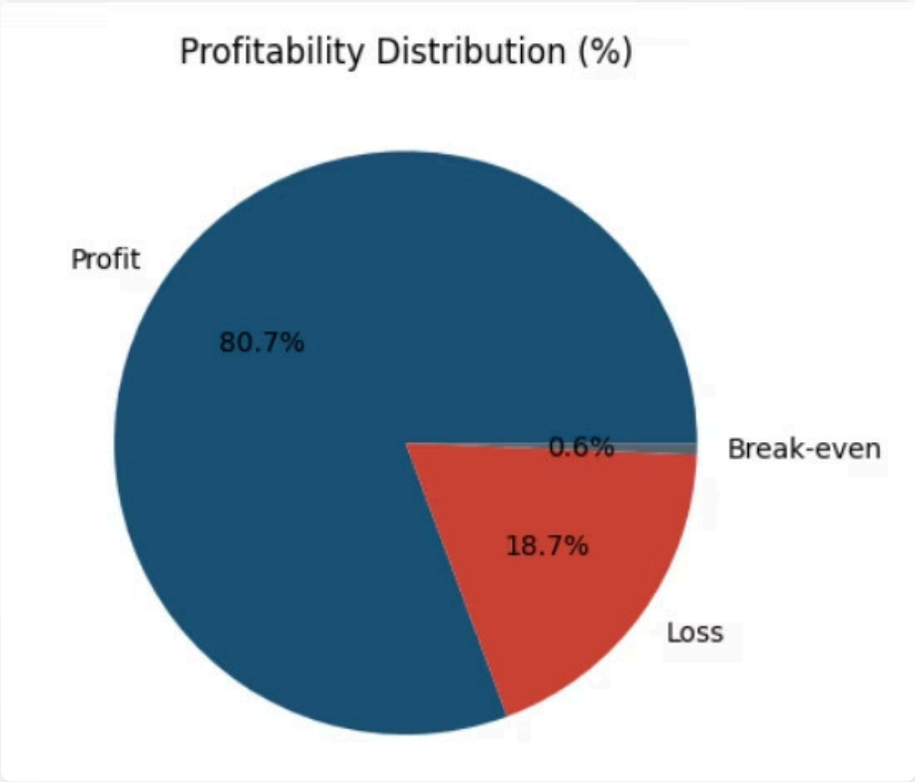
# Profitability Distribution Analysis

## Profit Distribution by Status

Analysis of 172,765 orders shows a clear profitability split. **80.7% of orders generate positive returns**, whilst 18.7% result in losses. However, the critical insight is that loss-making orders are heavily concentrated amongst delayed shipments.

This concentration suggests that operational friction, rather than product or pricing issues, is the primary driver of margin erosion. The financial risk is not that delayed orders lose more money individually, but that the volume of delays creates systemic profit pressure.




## Mean Profit vs. Delay Length

Data analysis reveals that **mean profit per order remains stable at approximately $22** regardless of delay length. This finding is crucial: we are not losing more money on "very late" orders compared to "slightly late" ones.

The financial risk is therefore a **volume problem**, not a severity problem. The business is losing $2.1M because too many orders are late, not because individual late orders are catastrophic. This insight shifts the solution focus from rescuing doomed orders to preventing delays at scale.

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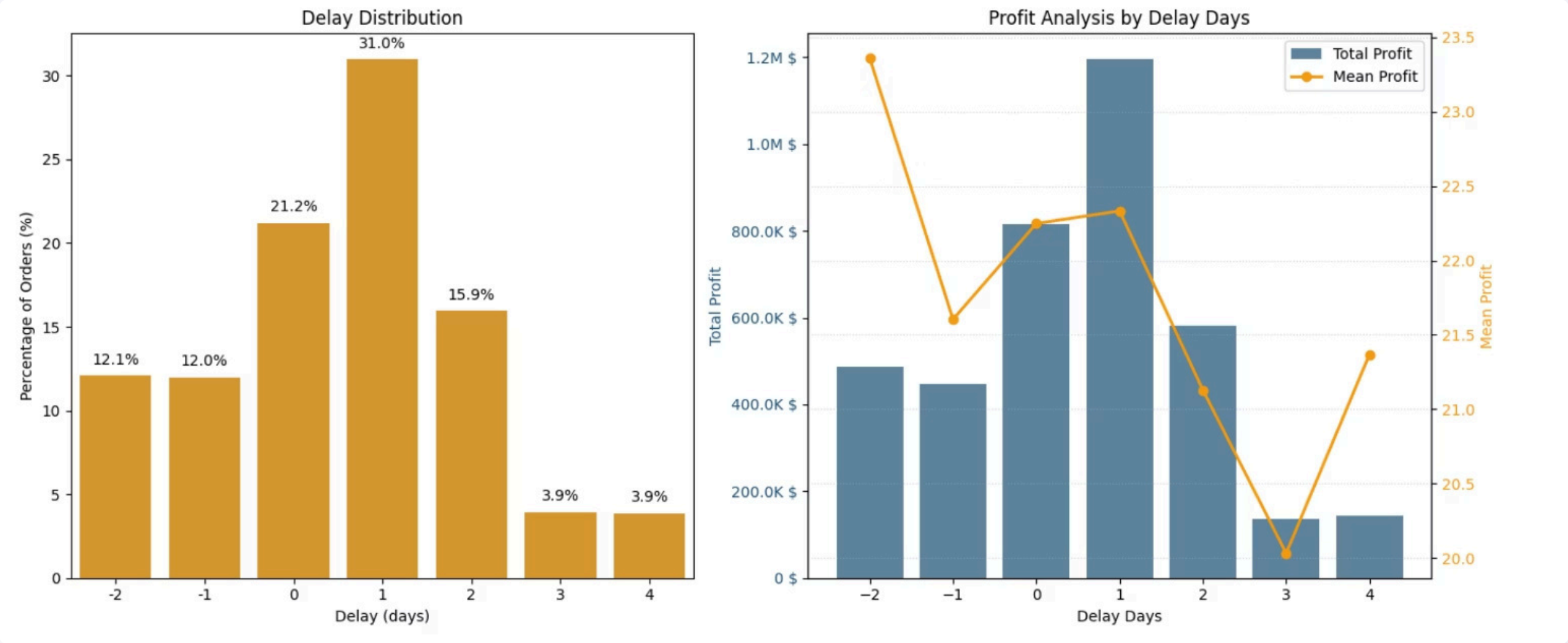
# Profitability vs Delivery Time Analysis

Profit vs. Delay: Data shows that mean profit per order remains stable at ~$22 regardless of the delay length. This suggests that the financial risk is a volume problem: we aren't losing more on "very late" orders, we are losing because too many orders are late.

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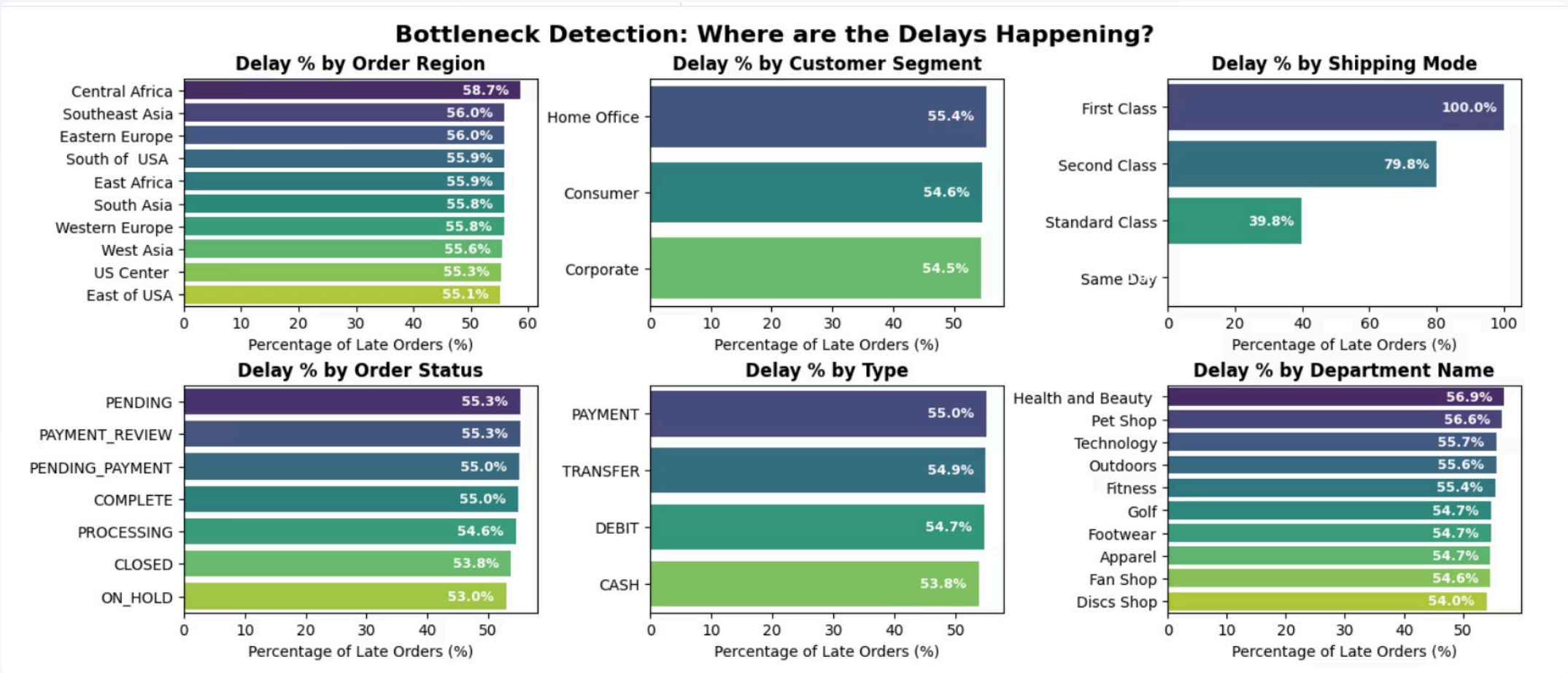
# Bottleneck Detection: The Dominant Lever

Analysis across six dimensions—shipping mode, region, customer segment, payment type, department, and time—revealed one dominant failure point that explains the systemic nature of the problem.




**100%**

### First Class Delay Rate

Every single premium order misses promised window

**79.8%**

### Second Class Delay Rate

Second-tier shipping also consistently fails

**55%**

### Standard Class Delay Rate

Base shipping performs better but still fails

**55-59%**

### Regional Variance

Consistent failure across all geographic regions

**54-55%**

### Customer Segment Variance

Identical delays regardless of customer type

**Critical Finding:** The near-identical delay rates across regions and customer segments prove this is not a local warehouse problem or a customer-specific issue. The failure is company-wide and systemic, pointing to flawed business logic rather than execution failure.

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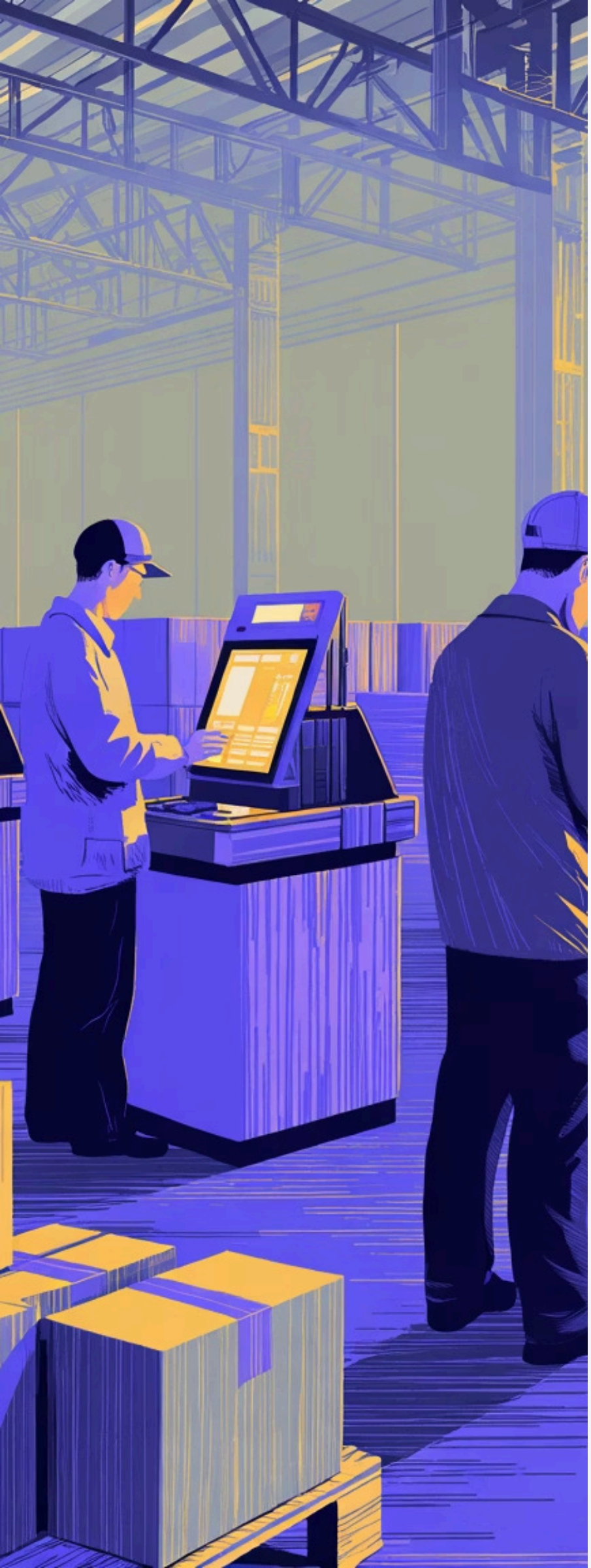
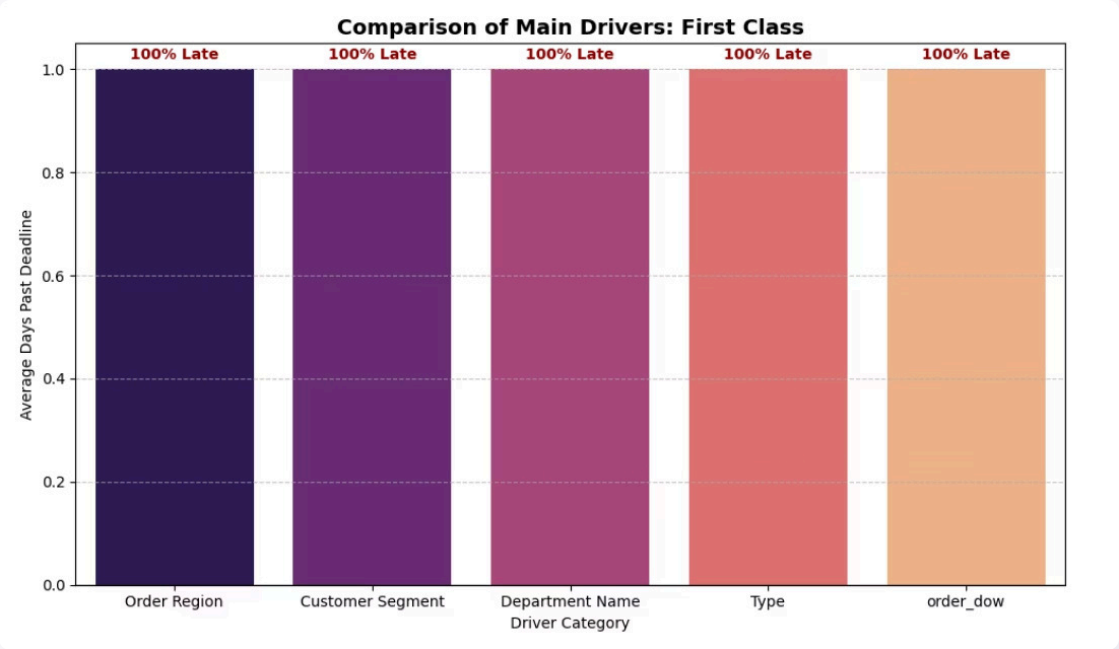


# Root Cause Analysis: The Impossible Promise

Deep-dive analysis into First Class shipping uncovered a fundamental systemic process flaw that explains the 100% failure rate.




### Hard-Coded Promise

The ERP system is hard-coded to promise a 1-day scheduled delivery window for First Class shipping, regardless of physical constraints

### Physical Reality

Actual processing time data confirms a constant 2.0 days regardless of payment type, department, or day of week

### Systemic Misalignment

The warehouse physically cannot meet the 1-day promise due to processing constraints and carrier handoff schedules

### 100% Failure

The 100% delay rate is caused by misaligned business logic in the ERP system, not poor employee performance or local execution issues

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# Time-Based Delay Patterns

## Monthly Peaks

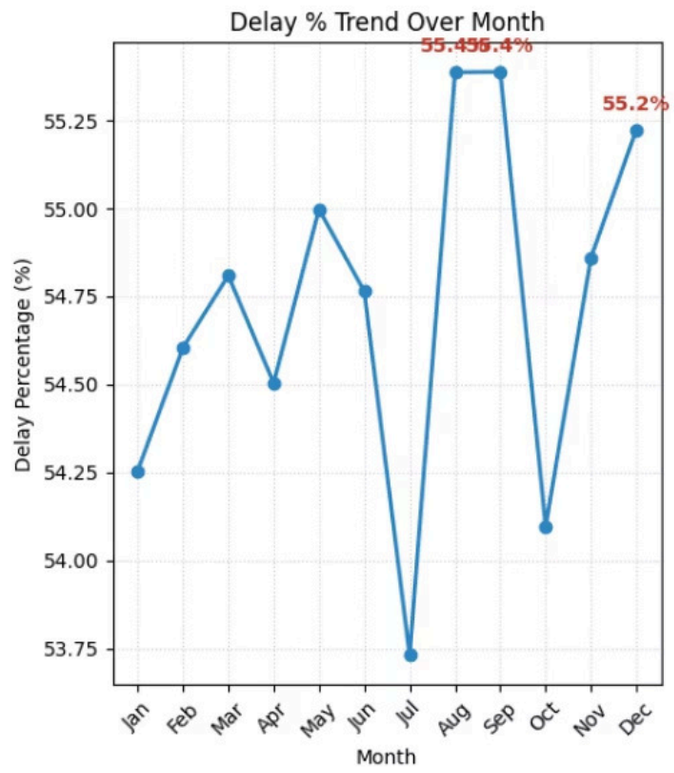
- August, September, December show 55%+ delay rates
- Correlates with holiday surges and promotional events
- Seasonal demand pressure amplifies existing system flaws

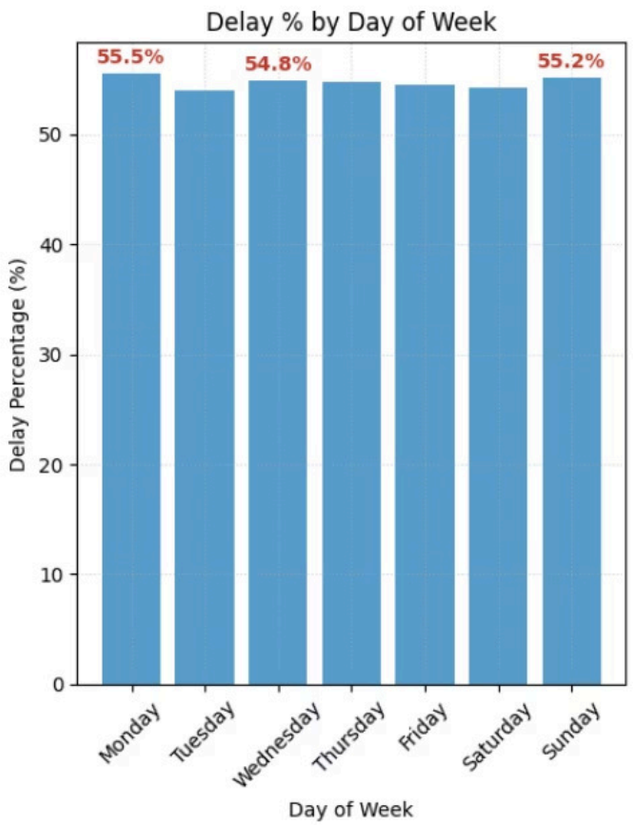
## Day-of-Week Variance

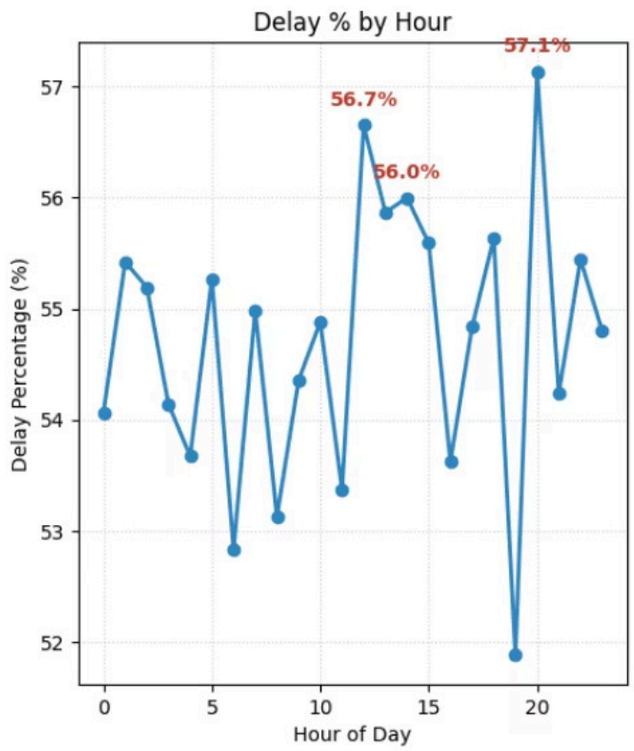
- Monday and Sunday see slight increases
- Variance is less than 1% across all days
- Indicates the issue is daily and constant, not weekly

## Hourly Spikes

- Orders placed at 9 PM (Hour 21) peak at 57.1% delays
- Likely missing final carrier cutoff for the day
- Late-evening surge requires staffing adjustment

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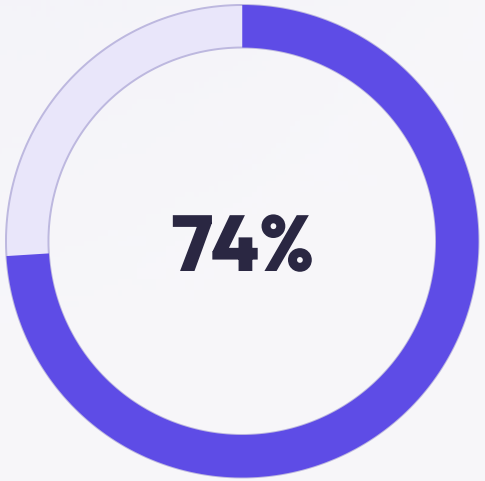
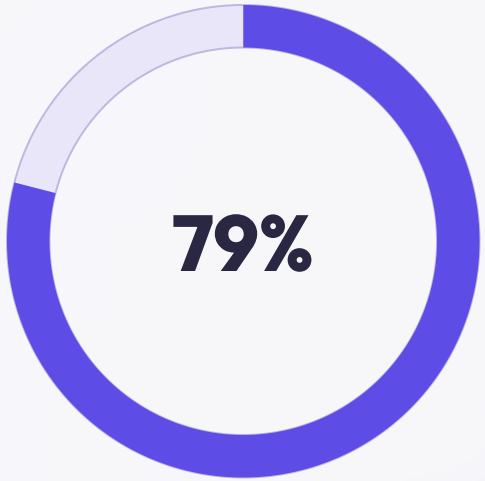
# Machine Learning Model Results

A Random Forest Classifier was trained using SMOTE (Synthetic Minority Oversampling Technique) to balance the classes and improve prediction for the minority late-delivery category.



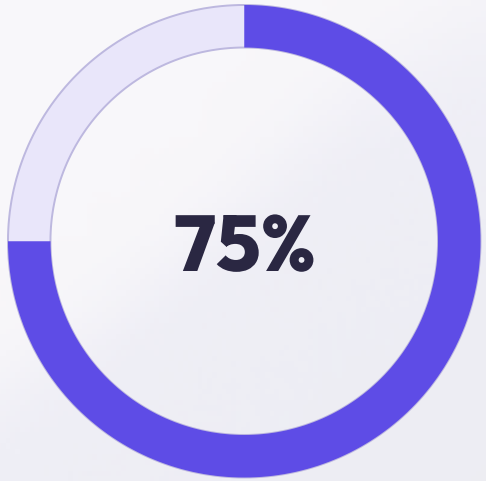

### Overall Accuracy

Correct predictions across both classes




### Precision (Class 1)

When model says "Late," it's correct 4 out of 5 times




### Recall (Class 1)

Model catches 75% of all actual late orders

## Model Interpretation

The model demonstrates robust performance for real-time risk flagging. With 79% precision, operations teams can confidently prioritise "High Risk" orders for expedited processing without wasting resources on false positives. The 75% recall rate ensures that three-quarters of all late orders are identified before dispatch.

## Deployment Priority

Integrate into Order Management System for real-time risk flagging

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# Strategic Recommendations & Conclusion

## 1

### Critical: Update ERP Logic

Immediately update First Class scheduled windows in the ERP system from 1 day to 2-3 days to match physical capability. This single change addresses the 100% failure rate at its source.

## 2

### High: Deploy Predictive Model

Integrate the Random Forest model into the Order Management System to flag "High Risk" orders for priority picking and expedited processing.

## 3

### Medium: Review Carrier Contracts

First and Second Class shipping significantly underperform Standard Class. Negotiate revised SLAs or consider carrier replacement.

## 4

### Medium: Staffing Adjustment

Increase warehouse staffing for the 9 PM shift to handle late-evening order surge and prevent missed carrier cutoffs.

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> **Target State:** Reduce late deliveries to under 30% within the next 12 months by fixing First Class logic flaws and deploying the predictive system. This shift protects $2.1M in profit and restores customer trust.

The 54.71% delay rate is not an unsolvable problem. It is the direct result of misaligned business logic that can be corrected through systematic process changes and predictive analytics deployment.

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